

Informe de experimento: pfagent

Generado el 2026-09-21 18:48 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio pfagent (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]
↪ low-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: pfagent
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 13:36
- Costo total: 1.6125 USD (40 items; costo medio 0.04031 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=567798
- Solved global: 60.0% (24/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo. **Esta reconstrucción es ilustrativa, no histórica:** usa el código de llm/prompts.py de HOY, así que si el texto del prompt cambió después de que esta corrida se generó, lo que se ve aquí no es lo que el modelo recibió. El registro histórico real es el hash por fila (system_prompt_hash, ver sección 1 / benchmarks/experiment_report.py::section_what_ran), fijado en el momento de la corrida y nunca recalculado.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) (reconstrucción con el código actual, no necesariamente histórica) — 1557 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given, unless the request itself states a
  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
  ↪ calling any tool.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del **scoreboard_per_case**), en los 4 grupos de **metricas_pfagent_definicion** **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u., solo sobre peticiones con formulación exacta −toda fila con herramientas que llega al solver con la formulación correcta da exactamente cero, ver benchmarks/scoring.py) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Escalated y Wrong (silent). Escalated y Wrong (silent) junto con solved_automously_rate (no la columna Solved de arriba, que puede solaparse con Escalated −una abstención de verificación puede caer sobre un item cuyo cálculo de fondo sí era correcto) suman 100 %: dónde termina cada petición −resuelta sola, entregada a una persona, o mal contestada sin aviso; solo cuenta como Escalated una arquitectura con mecanismo de traspaso real, ver benchmarks.scoring.escalation_check) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case N	Form.	V_MAE solved	V_MAE all	B_mean solved	B_mean all	Solved	Wrong-Escalated	Wrong-Infagged	Traceable	Time Tokens(s)	Claim	Wrong-Escalated	Wrong (silent)	\$
pfagent	case140	62.5 (25/40)	0	3.17e-04	−	8.26e-03	60.0 (24/40)	2.5 (1/40)	37.5	100.0	141957.37	−	2.5 (1/40)	37.5 (15/40)	1.6125

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
pfagent	80.0 (8/10)	90.0 (9/10)	40.0 (4/10)	40.0 (4/10)

Solved por dificultad (%, items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
pfagent	80.0 (8/10)	90.0 (9/10)	30.0 (3/10)	40.0 (4/10)

Verificación final (pfagent: V(x,c,z,y) aplicado a la respuesta final)

pass_first = pasó al primer intento; **pass_retry** = pasó tras el único reintento con el veredicto añadido; **abstained** = falló dos veces, respuesta forzada a fallo declarado sin números; **retry_mutation** = el reintento intentó una herramienta que muta la red (bloqueada, nunca ejecutada; cuenta como abstención) −ver “Integridad del reintento”abajo. N = items con **final_gate** activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained	retry_mutation
pfagent	40	90.0 (36/40)	7.5 (3/40)	2.5 (1/40)	0.0 (0/40)

Integridad del reintento

De 4 ítem(s) que llegaron a un reintento, 0 intentaron una herramienta que muta la red (bloqueada) en vez de corregir el reporte; se cuentan como abstención, no como “pasó”. El mensaje de reintento ya no sugiere la corrección específica (p. ej. “reconnect...”) y bloquea estructuralmente **modify_load**, **disconnect_line**, **reconnect_line**, **apply_remedial_action** y **load_case** mientras el reintento está en efecto.

Tipos de error de formulación (conteo de items)

Method	extra_arg_changes_result	missed_step	ok	wrong_id
pfagent	13	1	25	1

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5930 chars más}
gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers
- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if a ...[538 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers
- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any
- Voltage violations:
  - Bus 1: 1.060 pu, overvoltage
  - Bus 6: 1.070 pu, overvoltage
  - Bus 7: 1.057 pu, overvoltage
  - Bus 8: 1.090 pu, overvoltage
  - Bus 11: 1.064 pu, overvoltage
  - Bus 12: 1.054 pu, overvoltage
- Thermal violations: none
- The disconnected line 10-11 now has 0.0 MW flow, confirming it is out of service.

4. Suggested next step
If you want, I can now check which branch is most heavily loaded after this outage, run an N-1 contingency analysis, or recommend remedi ...
↳ [38 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_and_answer_ok); flags=-; error=-; tokens=5347;
↳ cost_usd=0.0173; wall=5.4 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-011-s0, case14, difficulty=ambiguous, seed=1971893210): Load case14 and set the active load at bus 6 to 14,700 kw.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 6, "p_mw": 14.7})
executed_calls: load_case({"case_name": "case14"}) -> get_status({}) -> modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
```

```

fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 263.13580543670497,
    ↪ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: get_status({})
  output: {"active_case": "case14", "network_info": {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11,
    ↪ "total_load_mw": 263.13580543670497, "total_gen_capacity_mw": 772.4000000000001, "has_last_result": false}
    gate: n/a (no es un resultado de flujo)
--- Ronda 3 ---
tool: modify_load({"bus_id": 6, "p_mw": 14.7, "q_mvar": 11.2})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.236323374534588, "is_violation": false,
    ↪ "violation_type": null}, {"bus_id": 3, "vm_p ...[6297 chars más]
    gate: PASS (reasons=, mismatch=6.47e-09 MW)
--- Ronda 4 ---
LLM: 1. I loaded IEEE case14 and set the active load at bus 6 to 14,700 kW, which solved successfully.

2. Key numbers:
- Total load: 267.145 MW
- Total generation: 281.819 MW
- Total losses: 14.674 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any:
- Voltage violations only; no thermal vio ...[248 chars más]
--- Respuesta final ---
1. I loaded IEEE case14 and set the active load at bus 6 to 14,700 kW, which solved successfully.

2. Key numbers:
- Total load: 267.145 MW
- Total generation: 281.819 MW
- Total losses: 14.674 MW
- Voltage range: 1.010 pu to 1.090 pu

3. Violations, if any:
- Voltage violations only; no thermal violations were reported.
- Overvoltage buses: 1, 6, 7, 8, 9, 10, 11, 12

4. Suggested next step:
- If you want, I can now check the most loaded branch, run an N-1 contingency analysis, or suggest remedial actions to reduce the
  ↪ overvoltage violations.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=9.52e-17; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-;
  ↪ error=-; tokens=6531; cost_usd=0.0194; wall=5.4 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```

request (case14-multistep-030-s0, case14, difficulty=multistep, seed=537958268): Load case14, set the active load at bus 14 to 17.2 MW,
  ↪ disconnect the line between bus 1 and bus 5, reconnect the line between bus 1 and bus 5, and finally run an N-1 contingency analysis
  ↪ ranked by min voltage and report the 8 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 14, "p_mw": 17.2}) -> disconnect_line({"from_bus": 1,
  ↪ "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8, "criteria": "min_voltage"})
executed_calls: load_case({"case_name": "case14"}) -> run_powerflow({}) -> modify_load({"bus_id": 14, "p_mw": 17.2, "q_mvar": 0}) ->
  ↪ disconnect_line({"from_bus": 1, "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8,
  ↪ "criteria": "min_voltage", "max_candidates": 0}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 271.9322468585065,
    ↪ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: run_powerflow({})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.373522845971019, "is_violation": false,
    ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[6356 chars más]
    gate: PASS (reasons=, mismatch=5.81e-09 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 14, "p_mw": 17.2, "q_mvar": 0})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.4514286734574515, "is_violation": false,
    ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "v ...[6408 chars más]
    gate: PASS (reasons=, mismatch=5.92e-09 MW)
--- Ronda 4 ---
tool: disconnect_line({"from_bus": 1, "to_bus": 5})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.382663773008685, "is_violation": false,
    ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0099999 ...[6332 chars más]
    gate: PASS (reasons=, mismatch=2.66e-08 MW)
--- Ronda 5 ---
tool: reconnect_line({"from_bus": 1, "to_bus": 5})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.4514286734574515, "is_violation": false,
    ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "v ...[6408 chars más]

```

```

gate: PASS (reasons=-, mismatch=5.92e-09 MW)
--- Ronda 6 ---
tool: run_n1_contingency({"top_k": 8, "criteria": "min_voltage", "max_candidates": 0})
output: {"plot_type": "n1_ranking", "figure_json":
  ↳ {"data": [{"cliponaxis": false, "hoverinfo": {"text": "", "hovernote": [{"\u003cbr\u003e\u003cbr\u003e: 1-5
  ↳ (line)\u003cbr\u003e\u003cbr\u003eConverged\u003cbr\u003e: True\u003cbr\u003e\u003cbr\u003eVoltage
  ↳ violations\u003cbr\u003e ...[19527 chars más]
gate: n/a (no es un resultado de flujo)
--- Ronda 7 ---
LLM: 1. **Summary:** I loaded IEEE case14, changed bus 14 active load to 17.2 MW, disconnected and then reconnected line 1-5, and finally
  ↳ ran an N-1 contingency analysis on the restored network ranked by minimum voltage for the 8 worst outages.

2. **Key numbers:**
  - Total load: **275.141429955464 M ...[1982 chars más]
--- Ronda 7 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.4514286734574515, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": -5.4514286734574515, "is_violation": false,
gate: PASS (reasons=-, mismatch=5.92e-09 MW)
--- Ronda 8 ---
LLM: 1. **Summary:** I verified the final restored case14 state after setting bus 14 active load to 17.2 MW and reconnecting line 1-5, and
  ↳ the N-1 analysis ranked by minimum voltage identifies outage **1-5** as the worst of the 8 requested outages.

2. **Key numbers:**
  - Total load: **275.1414299554 ...[1722 chars más]
--- Respuesta final ---
No numerical result is reported. Verification failed: the answer quotes 2 number(s) from a solve made before the last network change
  ↳ (reconnecting the branch between bus 1 and bus 5). This is a declared failure of the verification step, not a claim that the network
  ↳ failed to converge.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=1.62e-03; F_MAE=0.08; faithful=- %; solved=False (formulation);
  ↳ flags=abstained,abstained_on_solvable; error=-; tokens=99412; cost_usd=0.2674; wall=25.6 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V_MAE \geq 1e-4$ p.u. en 1 filas método×caso con solver: pfagent/case14 (3.17e-04).
- El gate verificó 58 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de pfagent: 60.0 % (24/40).
- Formulación exacta agregada sobre métodos con tools: 62.5 % (25/40).
- Formulación por dificultad: mejor parameterized 90.0 % (9/10), peor multistep 40.0 % (4/10).
- Errores de formulación más frecuentes: extra_arg_changes_result=13, missed_step=1, wrong_id=1.
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- 0 de 25 items con mutación de red reportaron estado obsoleto (stale_state).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 1.6125 USD; el método más caro fue pfagent con 1.6125 USD (100 %).
- Métodos con tools: 2.83 tool calls medias por item (máximo 7).

6. Archivos

Rutas absolutas bajo pfagent. report.rescored.json (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
  ↳ /power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/report.json [leído:
  ↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
  ↳ ids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard.rescored.json
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_n40/gpt-5.4/pfagent/traces

```